SAMAN FUYAL

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SUMMARY

Final-year Electrical & Electronics Engineering student at Kathmandu University seeking embedded systems and VLSI roles. Hands-on with ESP32/Arduino hardware, VHDL on FPGA, signal processing and wireless link analysis, supported by a Python developer certification.

EDUCATION

B.E. in Electrical & Electronics Engineering, Kathmandu University — *Expected Feb 2027*
CGPA: 3.5/4.0 (7 semesters completed) | Coursework: Embedded Systems, FPGA, Digital Signal Processing, Sensors, Electronics, Artificial Neural Networks
+2 Science, Greenland International College, Biratnagar — *2019/20 | GPA 3.1*
SEE, B.Y.M. Model Boarding School — *2021/22 | GPA 4.0*

PROJECTS

Performance Analysis of a LoRa-Based Communication System (ESP32, SX1278, Python) — *Ongoing*
- Simulated and field-tested a 433 MHz LoRa link for energy-efficient operation

Thermo-Oximeter, IoT Health Monitor (ESP32, MAX30100, MLX90614, MQTT) — *Third-year project*
- Portable SpO_2, heart-rate and temperature monitor with MQTT dashboard

Digital Clock on Artix-7 FPGA (VHDL, Vivado, Nexys A7) — *Jan 2026*
- Real-time HH:MM:SS clock with button debounce and 7-segment display

Traffic Signal Control using ANN and Fuzzy Logic (Python) — *Jul 2026 | Simulation*
- ANN waiting-time prediction with per-junction fuzzy signal control

4×10 Gbps WDM Optical System: PIN vs. APD (OptiSystem) — *Jul 2026 | Simulation*
- PIN and APD receiver comparison using Q-factor and BER

Other projects: Bidirectional Object Counter, 2025; IPv4 Subnetting (Cisco Packet Tracer), 2026; Line Following Robot, 2024

TECHNICAL SKILLS

Programming / HDL: C/C++, Python, MATLAB (strong), VHDL
Hardware: ESP32, Arduino, Artix-7 FPGA, I^2C, SPI, sensor interfacing, circuit prototyping and debugging
Tools: EasyEDA (schematic and PCB design), Simulink, Proteus, Xilinx Vivado, AutoCAD Electrical, Fusion 360, OptiSystem, Cisco Packet Tracer
Documentation: Microsoft Excel, Word, PowerPoint
Languages: English (fluent), Nepali (fluent)

CERTIFICATIONS AND TRAINING

Python Developer Certification, freeCodeCamp (~300 hours) — *Jun 2026*
Battery Management System, 24-hour seminar, Dept. of EEE, Kathmandu University — *Oct 2025*
AutoCAD Electrical Essentials, SourceCAD — *Sep 2025*
Home and Industrial Electrification, 3-day training, SEEE and Technical Training Center, KU

ACTIVITIES

- Technical Member, EEPEX (May 2026): led the Line Following Robot event, coordinated with teams for the Robosoccer event and gave training in the Home and Industrial Electrification program
- Volunteer, KU CAP 2025, Kathmandu University
- General Member, Society of Electrical and Electronics Engineers (SEEE) KU and Kathmandu University Robotics Club (KURC)